

EuCAIF guidelines and procedures

Sascha Caron, Elena Cuoco, Johan Messchendorp, Tilman Plehn, Christoph Weniger

EuCAIF management board 2025

Abstract

This document describes the EuCAIF guidelines and procedures. EuCAIF strengthens European research in fundamental physics by fostering cooperation and scientific exchange on machine learning and AI, connecting experts across physics and computational sciences, and training the next generation of researchers. It provides a community framework that stimulates new collaborations, supports coherent long-term planning, and helps develop a shared vision and direction.

Contents

1	Introduction	1
2	Membership and fellowships	2
3	Management Board	3
4	Advisory Board	4
5	EuCAIFCon	4
6	EuCAIF School	5
7	EuCAIF Working Groups	6
8	Funding and Community Support	6
9	Outreach and Web Presence	7

1 Introduction

The European Coalition for AI in Fundamental Physics (EuCAIF) operates as a collaborative, community-driven initiative and has been endorsed as a joint activity of ECFA, NuPECC, and APPEC within the JENAA framework*. Mission statement (as formulated in the JENAA mandate): *The “European Coalition for AI in Fundamental Physics (EuCAIF)” will establish a new European initiative for cooperation and scientific exchange on machine learning and AI in fundamental physics. This initiative will strengthen research efforts across Europe by bringing together AI/ML experts in particle physics, astroparticle physics, nuclear physics, and gravitational wave*

research with computational experts and AI/ML researchers.

This document defines the guidelines, rules, and procedures of EuCAIF, with the aim of ensuring stability while maintaining agility, inclusiveness, and a bottom-up, community-driven structure that serves the needs of the community. Their implementation is overseen by the EuCAIF Management Board. Amendments may be proposed by the Management Board and require approval by a simple majority vote of all EuCAIF fellows.

For long-term sustainability and effectiveness, a formal legal status may be beneficial. The Management Board may explore possible legal models, including an association, a not-for-profit foundation, or integration within existing European research infrastructures, in coordination with JENAA. No specific structure is proposed at this stage. Amendments concerning the legal status of EuCAIF require approval by a two-thirds majority of EuCAIF fellows.

Publications and Acknowledgements

Publications arising from EuCAIF activities remain the responsibility of the contributing authors and groups; authorship is self-organized within the relevant teams. When appropriate, publications are encouraged to acknowledge EuCAIF’s role. In cases of uncertainty regarding authorship or the use of the EuCAIF name, the Management Board is expected to be consulted.

*<http://nupecc.org/jenaa/>

2 Membership and fellowships

EuCAIF is establishing a network of researchers across European institutions to bridge fundamental physics and AI research.[†] The network connects leading scientists who visibly advance both fields, provide scientific leadership to the community in Europe, and train the next generation to drive this transformative work forward.

There are currently three levels of involvement in EuCAIF: EuCAIF members, EuCAIF junior fellows, and EuCAIF fellows.

EuCAIF member

Anyone may become a *EuCAIF member* by subscribing to the eucaif-info mailing list. Membership is intended as an open and lightweight way to stay connected with EuCAIF.

EuCAIF fellows

EuCAIF fellows guide the strategic direction of EuCAIF, contribute actively to the implementation of its activities, and participate in regular meetings to take key decisions, including the election of Management Board members.

Scientists holding permanent scientific positions at European institutions, or equivalent long-term appointments, with a strong track record and high visibility in AI and fundamental physics may apply for a (renewable) five-year EuCAIF fellowship. Candidates are expected to meet the following criteria:

1. Internationally recognized, high-impact publications in their research field;
2. High-impact publications on AI methods relevant to fundamental physics, in either AI or physics venues;
3. A demonstrable track record of training young scientists advancing AI in fundamental physics;
4. Active and leading contributions to EuCAIF activities as a member, or a clear plan

for such contributions presented in the application.

Exceptional candidates who do not yet meet all criteria may still be considered.

Fellowship applications should consist of a short expression of interest (maximum 0.5 page, excluding references) outlining how the applicant envisions contributing to EuCAIF and how they have contributed in the past to AI in fundamental physics. Applications are reviewed by three independent referees selected by the Management Board: one from the candidate's field, one from a related field, and one Management Board member. Final decisions are made by majority vote of all Management Board members.

Fellowships are granted for five years. Renewal is normally granted to fellows who demonstrate sustained contributions to EuCAIF, for example through regular meeting attendance and active participation in activities. Fellows who show limited or irregular engagement may be contacted by the Management Board to confirm their continued interest before renewal is decided.

EuCAIF junior fellows

EuCAIF junior fellows are scientists at advanced early-career stages (typically tenure-track, or equivalent) who do not yet hold a tenured or equivalent permanent position at a European institution, who are on a trajectory toward such positions, and who demonstrate strong engagement with EuCAIF's mission.

The same four criteria as for EuCAIF fellows apply, but they are assessed relative to career stage. In particular, evidence of potential may substitute for a full track record in areas such as training young scientists or providing scientific leadership. EuCAIF junior fellows automatically transition to EuCAIF fellows upon obtaining a tenured or equivalent permanent position at a European institution. Applications for junior fellowships follow the same procedure as fellowship applications and require a brief expression of in-

[†]As of 2025, EuCAIF includes members and activities across the physics domains of ECFA (particle physics), NuPECC (nuclear physics) and APPEC (astroparticle physics, cosmology, gravitational waves). Membership is open to individuals from these communities and from related fields, including astronomy, AI research, and industry, where activities align with EuCAIF's mission. EuCAIF follows the broad definition of "Europe" used by JENAA and its supporting organisations (ECFA, NuPECC, and APPEC), which serves as the reference for eligibility.

terest (maximum 0.5 page, excluding references), assessed relative to career stage.

3 Management Board

The Management Board is the executive body responsible for the coordination and strategic implementation of EuCAIF activities. It consists of five EuCAIF Fellows elected for fixed terms, together with one Junior Fellow representative.

Tasks of the Management Board

The Management Board has the following core responsibilities:

1. **Strategic oversight and governance**
To define, maintain, and regularly update the internal rules, procedures, and governance structures of EuCAIF, ensuring alignment with the objectives set out by EuCAIF and its supporting organizations (in particular ECFA, NuPECC, and APPEC).
2. **Organisation of EuCAIFCon and other community activities**
To plan, coordinate, and oversee the annual EuCAIF Conference (EuCAIFCon), including venue selection, scientific programme, invited speakers, and logistical arrangements. To support or initiate additional workshops, training schools, and community meetings that advance the mission of EuCAIF, and to actively promote and encourage community-driven proposals for these activities.
3. **Funding and resource support**
To develop and promote strategies for securing resources that support EuCAIF's mission, including community-building instruments such as COST actions or similar programmes. To stimulate and endorse joint proposals by EuCAIF members to national and international funding agencies, and to provide input on the allocation of resources (e.g. for travel grants, events, working groups or schools) in coordination with

the relevant funding structures.

4. **Appointments and decision-making procedures**

To organise and oversee transparent internal procedures for appointments and votes of strategic importance, including the selection of Fellows, the confirmation of working groups, and changes to governance rules. The Management Board shall ensure fairness, clear communication of procedures and eligibility, and alignment of all outcomes with EuCAIF's mission and commitment to diversity and scientific excellence.

5. **Coordination with the Advisory Board and Fellows**

To serve as the primary liaison between the EuCAIF Advisory Board and the Fellows, ensuring that the Advisory Board's guidance is considered in EuCAIF's activities and that information flows transparently in both directions.

6. **Reporting and communication**

To maintain clear records of EuCAIF's decisions and activities, and to ensure timely communication of key developments and opportunities to Fellows, members, and external stakeholders.

The Management Board may self-organise its internal roles and responsibilities, distribute coordination tasks among its members, and designate one or more coordinators or a chair as needed; these roles and primary contact points should be made visible on the EuCAIF webpage to ensure transparency and accessibility.

Board Member Selection Procedure

Management Board members serve fixed terms of two years. Elections are held annually, with two or three of the five Board positions subject to election each year, such that half of the Board is renewed per year on a staggered basis.[‡] Term limits and staggered rotation are designed to ensure continuity of experience within the Management Board while preventing full turnover in any single election cycle. Mid-term vacancies are not

[‡]Transitional arrangements for the initial implementation of the staggered election cycle shall be defined by the Management Board and communicated to the Fellows in advance; these arrangements apply only to the first election under the new system.

backfilled. Immediate re-election is excluded, but Fellows may stand again after one full term off the Board.

In addition to these members, the Management Board includes one Junior Fellow representative. This representative is elected annually by and from the Junior Fellows for a one-year term and serves as a full voting member of the Board. The Junior Fellow representative may stand for re-election once for a consecutive term.

A Nomination and Election Committee (NEC) of at least five members is proposed by the outgoing Board and confirmed by a simple majority of Fellows. The NEC organises nominations and voting, and ensures fairness and transparency for all Board elections. Its composition should ideally include a mix of Fellows, Junior Fellows, and Advisory Board members, and exclude current members of the Management Board. The NEC issues an open call for nominations to all Fellows, specifying the submission process and deadline. Members of the NEC may not stand as candidates in the election they oversee.

Candidates for the five regular Board positions must be current Fellows and may be nominated by others or may self-nominate. Each candidate shall submit a brief vision statement (maximum one page) outlining their intended contributions. In the election for the five regular Board positions, each Fellow and Junior Fellow may cast up to five votes, one for each preferred candidate. The Junior Fellow representative is elected in a separate ballot in which only Junior Fellows may vote.

The top five candidates for the regular Board positions are elected, subject to diversity criteria: representation across the scientific domains corresponding to ECFA, NuPECC, and APPEC (particle physics, nuclear physics, and astroparticle physics / cosmology / gravitational-wave research), where possible, and at least two different genders. If unmet, the NEC applies a lowest-out/highest-in adjustment until criteria are satisfied. Ties are resolved by maximising total votes; residual ambiguity is settled by NEC majority.

Before the ballot is issued, the NEC confirms that all nominees meet the eligibility requirements and are willing to serve if elected; only confirmed can-

didates appear on the ballot.

After the election, the NEC contacts the candidates selected under the above procedure to re-confirm their willingness to serve. If a selected candidate declines, the NEC replaces them by the next highest-ranked candidate, reapplying the diversity criteria as needed. Once all positions are accepted, the NEC announces the final list of elected Board members and, where applicable, notes whether diversity adjustments were applied. Detailed vote counts remain confidential to the NEC but may be audited upon request by Fellows to ensure compliance with the established rules.

4 Advisory Board

The Advisory Board provides strategic advice to EuCAIF and supports its long-term vision through expert guidance across science, policy, and infrastructure. Advisory Board members have no executive or voting role in EuCAIF governance, but serve as a resource for the Fellows and the Management Board.

Appointments

Members of the Advisory Board are external to the EuCAIF fellowship. They are usually senior figures selected for their expertise, leadership, and ability to contribute to EuCAIF's strategic direction. Fellows may suggest names, but appointments are made at the discretion of the Management Board, which decides by majority vote. Proposals and deliberations remain confidential. Appointments are open-ended but may be reviewed periodically to ensure continuity, balance, and active engagement. The composition of the Advisory Board may also be reviewed periodically by the Management Board to ensure appropriate diversity and continued active engagement. The size of the Advisory Board is determined by the Management Board to ensure broad and balanced representation.

5 EuCAIFCon

EuCAIFCon is the flagship annual meeting of EuCAIF, providing a forum for scientific exchange

and community building at the interface of fundamental physics and AI. The conference is held once per year as a summer event to accommodate the academic calendar and maximise participation from European institutions.

Site Selection

Each year, the Management Board issues a call for nominations for the following year’s EuCAIF-Con venue, typically around eighteen months before the event in question, and normally at least six months before the upcoming conference. Fellows may nominate their home institutions or other suitable venues within Europe. The call for nominations is open to all Fellows and Junior Fellows, and should follow transparent procedures. Proposals should include information on the venue and capacity, the local organising committee, budget considerations, available facilities and accommodation, accessibility, and how continuity with the established EuCAIFCon structure will be maintained (e.g. plenary and parallel sessions, poster sessions, networking opportunities). Venues should be able to host a meeting of comparable scale to recent EuCAIFCon editions.

Voting and Decision

The Management Board organises a secure online vote listing all valid nominations. All Fellows and Junior Fellows may cast one anonymous vote. The site receiving the most votes is selected, subject to confirmation by the Management Board. The decision is normally finalised during, or shortly before, each EuCAIFCon, so that the venue for the following year can be announced at the close of the current meeting.

Responsibilities

The Management Board has overall responsibility for EuCAIFCon. A Local Organising Committee (LOC), proposed as part of the site nomination, manages logistics under the oversight of the Board. A Scientific Programme Committee (SPC), formed jointly by the Board, Fellows, and LOC, coordinates the scientific programme. Where possible, the SPC should include members of the Management Board as well as other Fellows and Junior Fellows. The SPC ensures alignment with EuCAIF’s mission, transparent procedures,

and continuity with past events, while allowing room for innovation.

6 EuCAIF School

EuCAIF Schools are planned as flagship training initiatives that equip early-career researchers with the theoretical foundations, practical skills, and interdisciplinary mindset needed to advance artificial intelligence in fundamental physics. Their objectives are to provide high-quality training at the interface of AI and physics, to foster collaboration across disciplines such as physics, computer science, and data science, and to support the development of the next generation of leaders in AI for science.

Organisation

The Management Board issues calls for proposals for EuCAIF Schools and selects among them; in addition, proposals for EuCAIF Schools may be submitted at any time and considered independently of a formal call. Proposals are typically submitted by Fellows or groups of members and should demonstrate scientific excellence, organisational feasibility, and alignment with EuCAIF’s mission. Scientific coordination is the responsibility of a dedicated organising committee, which may include Fellows, invited lecturers, and representatives of partner institutions. EuCAIF strives to hold at least one school per year. Approval of a EuCAIF School is granted for a single edition; subsequent editions require a new proposal and re-approval by the Management Board.

Format and Participation

Each school is expected to combine lectures by leading experts, hands-on tutorials and coding labs, collaborative project sessions, and opportunities for career development, with explicit emphasis on interdisciplinary perspectives bridging physics, computer science, and data science, as well as open science practices and discussions on the responsible use of AI. Participation is open to Master’s students, PhD candidates, and early postdocs. Admission is competitive and based on academic merit, motivation, and the relevance of the applicant’s background to EuCAIF’s goals, while aiming to ensure diversity across dis-

ciplines, career stages, institutions, and demographics among participants.

7 EuCAIF Working Groups

EuCAIF Working Groups (WGs) are thematic task forces that focus on specific scientific, technical, or strategic areas central to EuCAIF’s mission. They provide a flexible structure for community collaboration, the development of tools and standards, and the pursuit of strategic initiatives.

Scope and Responsibilities

WGs operate within a clearly defined scope. Their responsibilities may include developing benchmarks, datasets, tools, or methodologies; coordinating efforts around specific challenges or applications; producing white papers, reports, or roadmaps; and organising topical workshops, tutorials, or collaborative projects. Each WG is led by one or two coordinators, who are responsible for guiding its activities and reporting progress to the community and the Management Board. WGs are expected to hold dedicated sessions and report their progress at EuCAIFCon. Additional WG meetings outside EuCAIFCon are encouraged. WGs are expected to self-organize and identify a WG leader, who reports on the WG progress at least once per year, typically through a dedicated session or contribution at EuCAIFCon.

Opening and Tasks

New WGs may be proposed by any EuCAIF Fellow or Member. Proposals must outline the motivation, scope, expected impact, initial membership, and a plan of tasks and deliverables, including objectives, milestones, and a timeline for achieving them. The Management Board evaluates each proposal and submits it to the Fellows for confirmation by simple majority vote. Approved WGs are established with the expectation of completing their work within the proposed timeline, though extensions may be granted if justified.

Closing

A WG may be closed (i) by decision of the Management Board, if it has completed its objectives, become inactive for an extended period, or is no longer aligned with EuCAIF’s strategic priorities; or (ii) upon proposal by the WG coordinators and agreement of the Management Board. In cases of doubt, the Management Board may consult the Fellows by vote before making a final decision.

8 Funding and Community Support

A central role of EuCAIF is to promote the relevance and impact of artificial intelligence in fundamental science at the European level, including through the support and coordination of national and international funding streams and proposals that advance AI for fundamental physics. EuCAIF provides coordination, visibility, and community alignment for emerging initiatives, and helps build competitive, collaborative funding strategies.

European Strategy and Policy Engagement

In pursuing this role, EuCAIF may engage with European institutions, research infrastructures, funding bodies, and relevant strategic initiatives, serving as a natural interlocutor for European strategy processes and policy discussions related to AI in science. Drawing on the combined perspectives of the communities represented by ECFA, NuPECC, and APPEC, such engagement is intended to enhance the visibility of fundamental physics within broader European AI strategies and to ensure that the needs and opportunities of these communities are appropriately represented.

Coordination and Matchmaking for Funding Opportunities

The Management Board tracks funding opportunities at the European and international level and informs the community about areas where coordination may be beneficial. EuCAIF facilitates match-making between potential coordinators and participants, for example through internal calls for expressions of interest. Decisions on

the composition and leadership of proposals remain with the proposers, though when the Management Board is involved in suggesting key roles, it seeks to ensure broad participation, scientific balance, and rotation of leadership opportunities.

Formal Support and Endorsement

EuCAIF may support proposal development by providing letters of support, strategic endorsement, access to white papers, benchmarks, or data repositories from Working Groups, and platforms for coordination and matchmaking. Proposals that wish to use the EuCAIF name or present themselves as formally endorsed by EuCAIF must seek approval from the Management Board. Endorsement is decided by the Board through a simple majority vote, with members recusing themselves from deliberation and voting in cases of conflict of interest, in line with standard good-governance practice. Endorsed proposals are communicated to all Fellows, and accepted proposals will also be listed on the EuCAIF website.

EuCAIF-led Funding Proposals and Community-Building

EuCAIF's primary role in funding activities is to support and coordinate community-building. EuCAIF itself may apply for or coordinate funding instruments that strengthen shared community infrastructure, such as workshops, schools, or networking schemes (e.g. COST actions and comparable coordination, infrastructure, and networking schemes). The Management Board may coordinate such initiatives and develop joint strategies to secure resources for EuCAIF's operations and shared infrastructure. The scope of EuCAIF-led proposals may evolve over time in response to community needs.

In cases where EuCAIF jointly applies for large-scale community-building and research grants that require a limited consortium size (e.g. Horizon Europe RAISE or comparable instruments), participation may be restricted to a subset of EuCAIF members or affiliated institutions. For each such proposal, the Management Board appoints a dedicated proposal writing committee with a clear mandate, timeline, and scope. The writing committee is responsible for developing the pro-

posal concept and selecting participating institutions and consortium partners based on transparent criteria aligned with the objectives and constraints of the funding call. All proposals submitted under the EuCAIF name must include clear mechanisms, in line with the objectives and constraints of the funding call, by which the resulting resources support the broader EuCAIF community, for example through conference support, fellowship or training positions, travel grants, PhD schools, or shared infrastructure.

Sustainability

EuCAIF encourages all proposals developed within its community to include elements that strengthen long-term community benefit, such as shared infrastructure, datasets, and tools; training and outreach aligned with EuCAIF Schools and EuCAIFCon; and adherence to FAIR and open science principles. Where possible, proposals are encouraged to allocate resources that contribute to EuCAIF's activities and infrastructure.

9 Outreach and Web Presence

The EuCAIF webpage serves as the central information hub for the EuCAIF community and for the wider European research community, including partner organisations such as ECFA, NuPECC, and APPEC. It provides up-to-date information on EuCAIF activities, governance, working groups, and events, and is designed to be openly accessible and welcoming to newcomers. In its development, attention is given to accessibility, clarity, and adherence to open science and FAIR principles.

Together with other associated communication channels, the webpage serves as the primary instrument for EuCAIF's outreach, public communication, and engagement with the broader scientific community, European institutions, and other stakeholders. Through these channels, EuCAIF aims to communicate its mission, activities, and positions clearly and accessibly, and to support dialogue with partner organisations and institutions.

Purpose, Content, and Visibility

The webpage presents general information about EuCAIF, its mission, and organisational structure. It provides details on the Management Board, Advisory Board, Fellows, and Working Groups, as well as information on governance structures, election procedures, and EuCAIF-endorsed activities, and announces events such as EuCAIFCon, the EuCAIF School, and topical workshops. The webpage should clearly indicate the current distribution of Management Board roles and responsibilities, including designated coordinators or chairs, together with appropriate contact points.

Public deliverables, including white papers, benchmarks, and software tools, are made available, along with guidelines for joining EuCAIF or proposing new initiatives. In addition, the webpage highlights news, funding opportunities, and job openings of interest to the community,

and is foreseen to serve as a lightweight meta-repository of relevant existing resources from the JENAA communities (e.g. links to connected working groups, living reviews, datasets, and shared tools). The webpage also serves as a visible point of contact for external stakeholders, including other AI initiatives, research infrastructures, policy bodies, and industry partners interested in engaging with EuCAIF.

Maintenance, Accuracy, and Updates

The EuCAIF webpage is maintained under the supervision of the Management Board, with technical support delegated to designated Fellows or partner institutions. Content updates are coordinated with Working Group leads and event organisers to ensure accuracy and timeliness. In particular, the lists of Fellows, Junior Fellows, and active Working Groups should be kept up to date.